Exploiting Spack for MPICH's CI

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Speakers

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- Mathematics and Computer Science

MPICH

MPICH is a leading MPI implementation and a large open-source project.

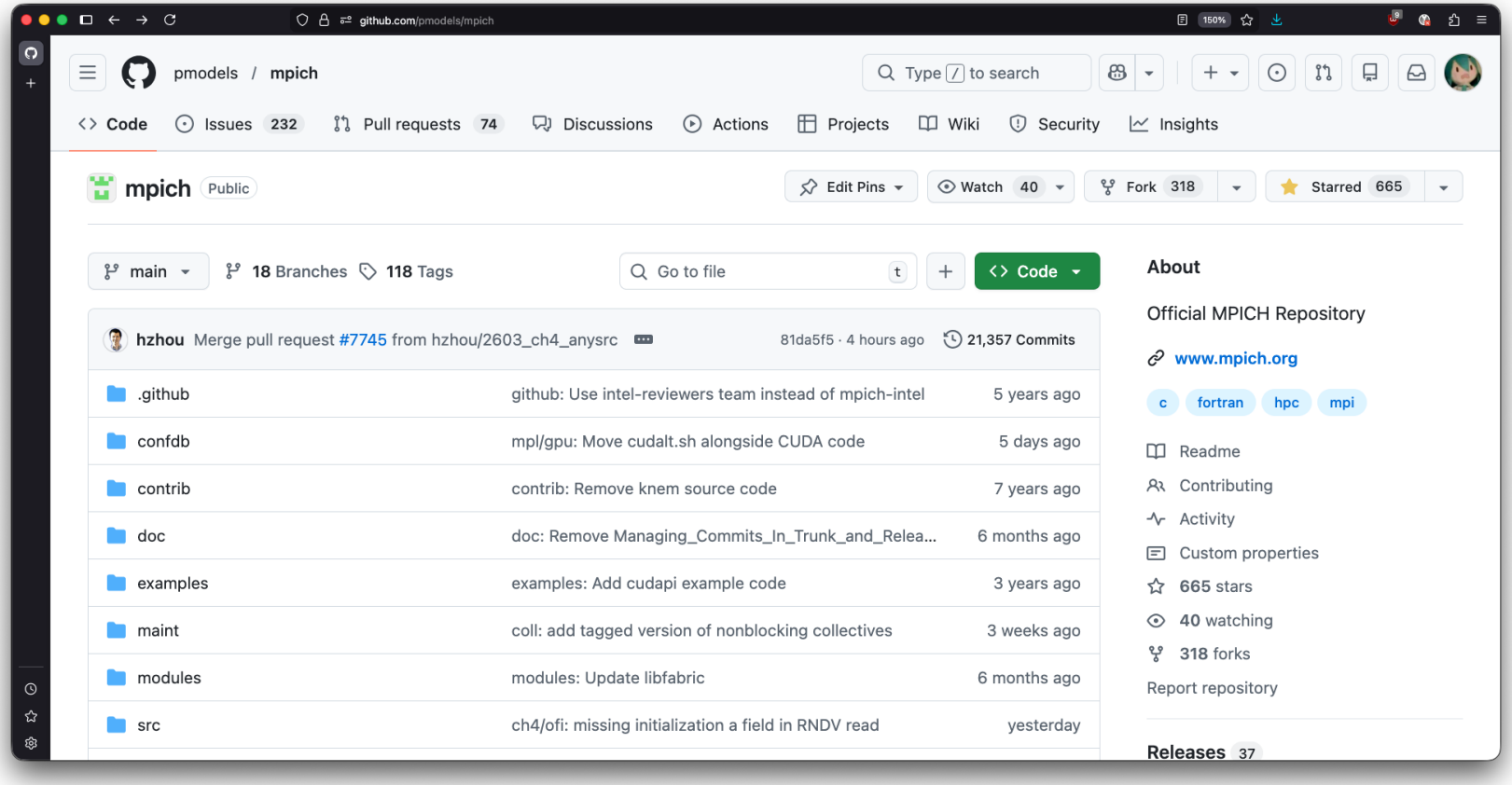
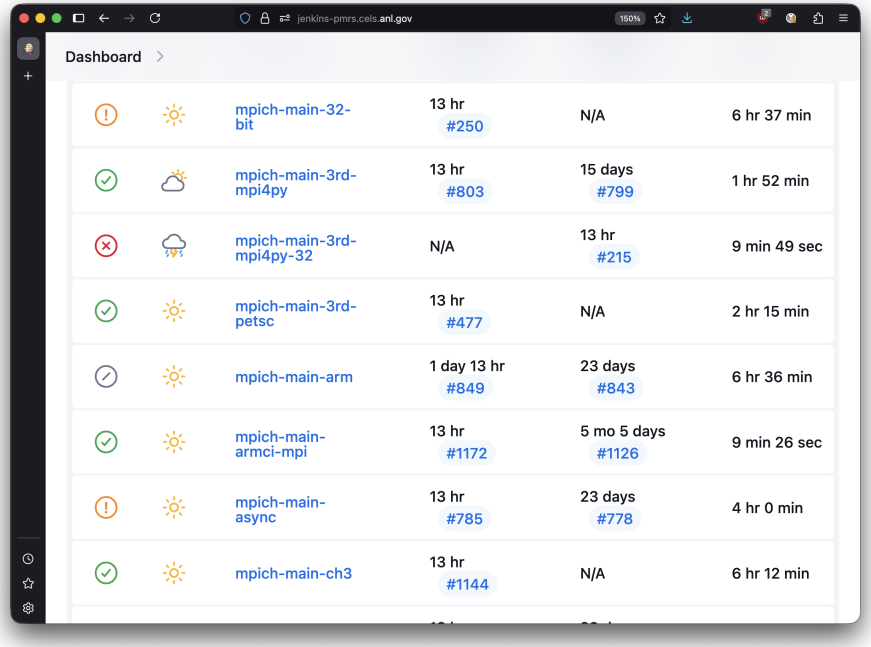


Figure: MPICH GitHub[1]

[1] github.com/pmodels/mpich

Our Maintainer Workflow



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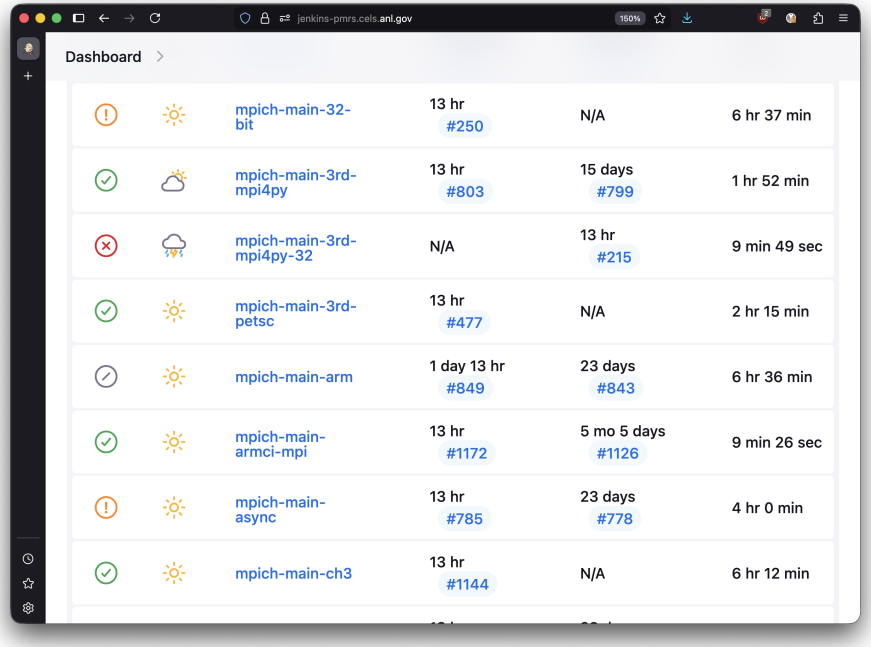

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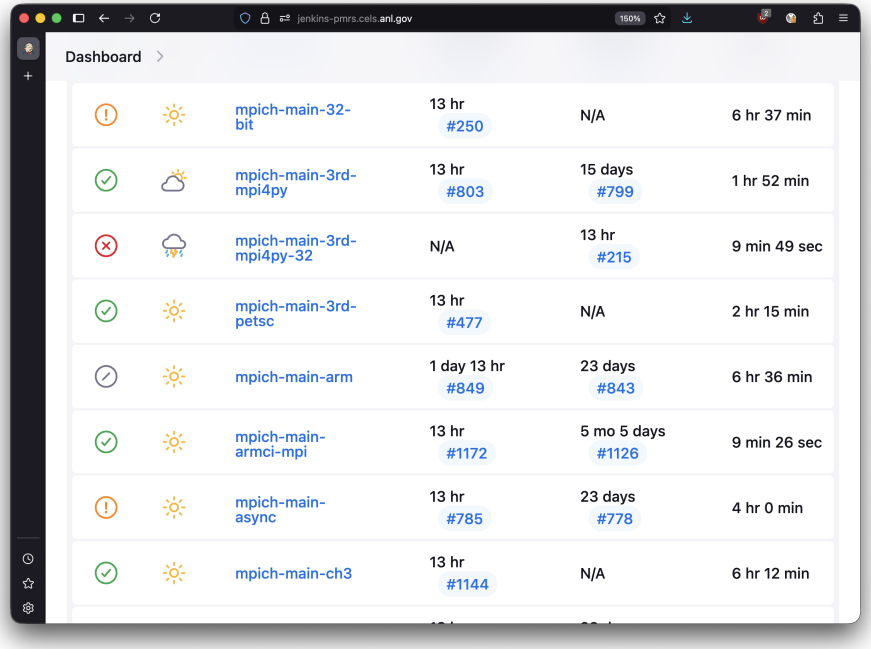

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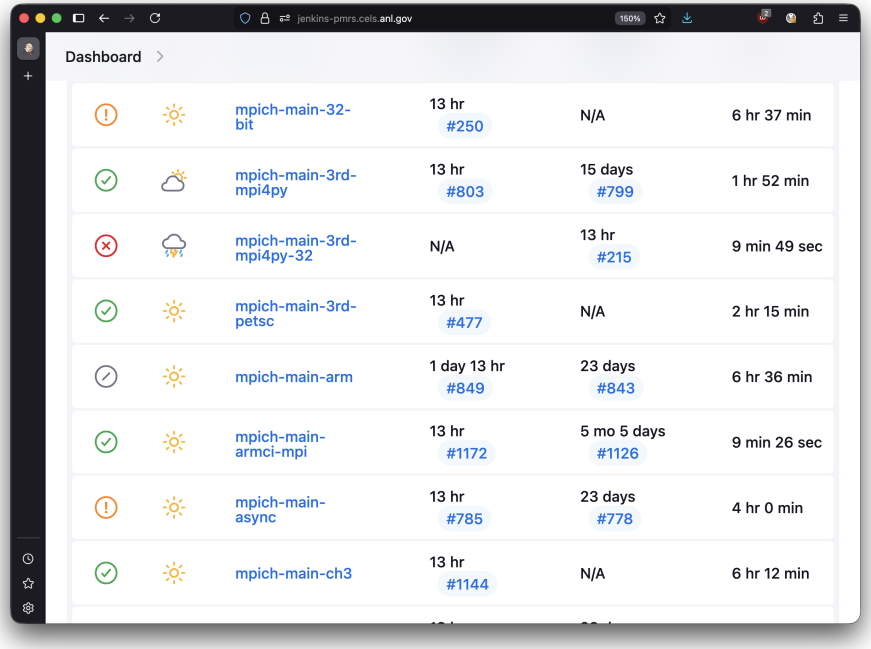

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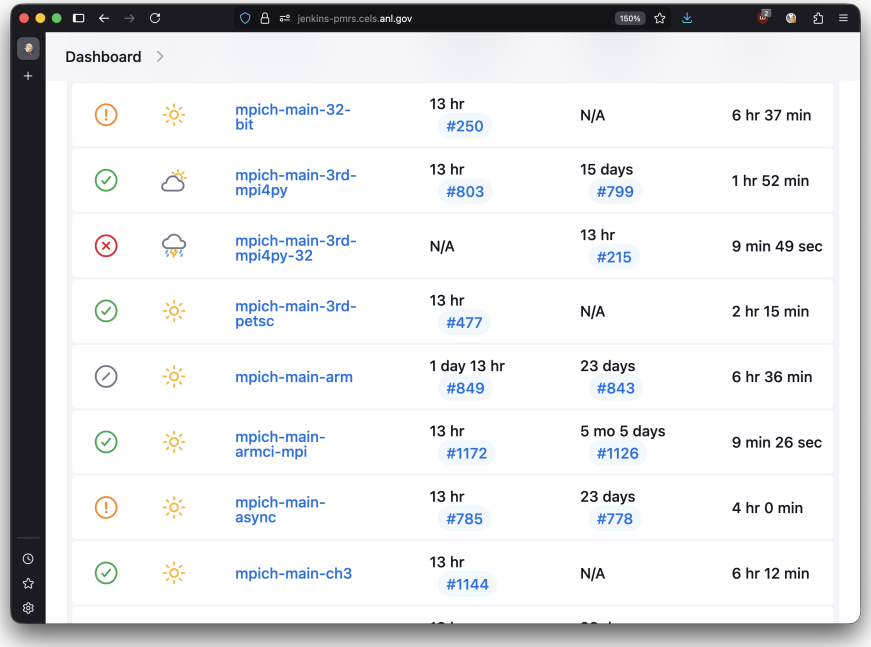

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  - etc...
- All PRs test against CI
- `main` validated nightly

Our Maintainer Workflow

$$\begin{aligned}\text{Nightly Jobs} &\approx 300 \text{ matrix configurations}\\ &\mapsto N \times M \times \ldots \text{ combinations}\\ &\Rightarrow \textbf{1500} \text{ runs } \underline{\text{nightly}}\end{aligned}$$

Our Maintainer Workflow

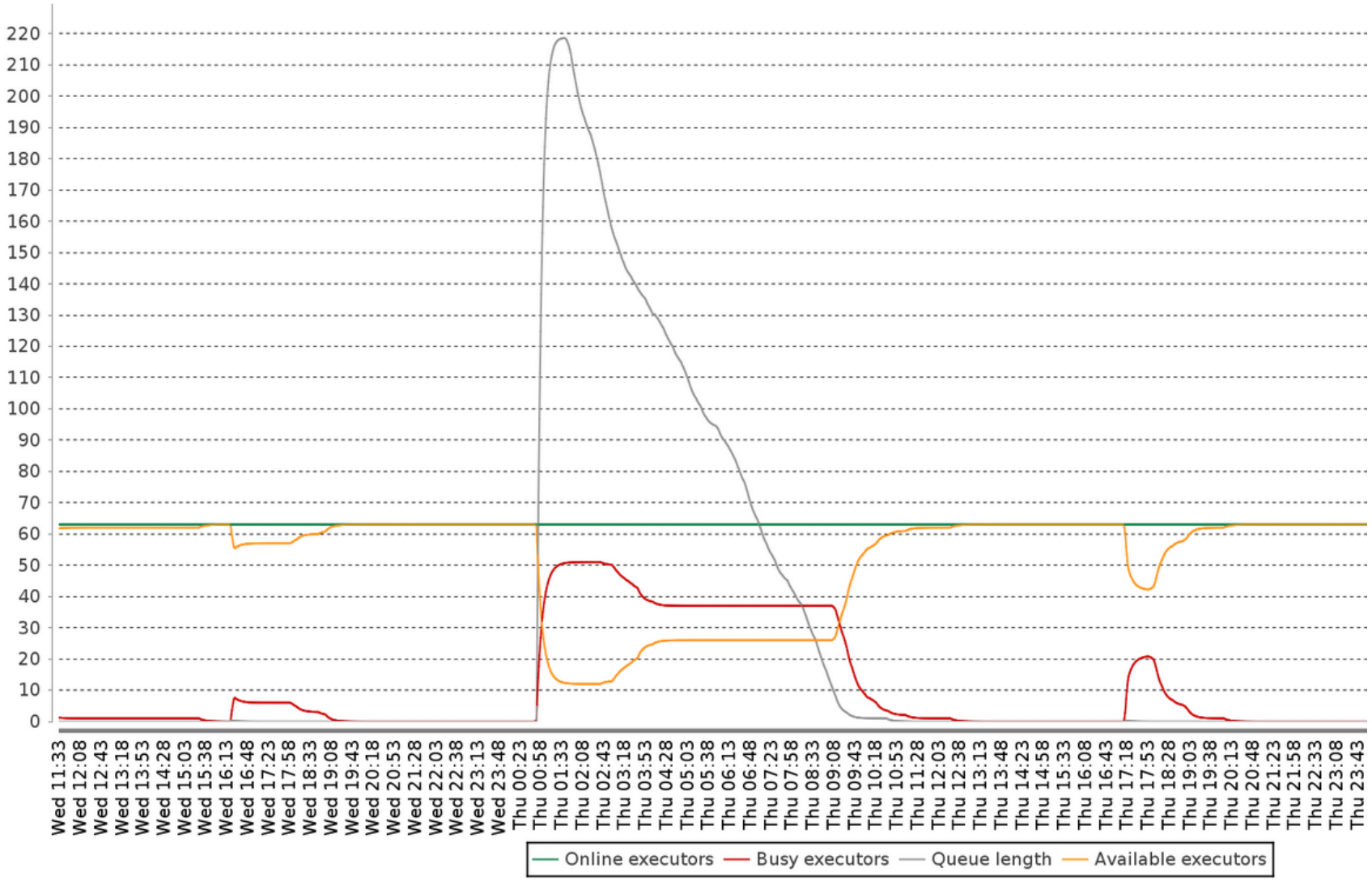

Figure: Jenkins Load Usage Statistics

Qualms

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$\sim$1 hour 30 minutes

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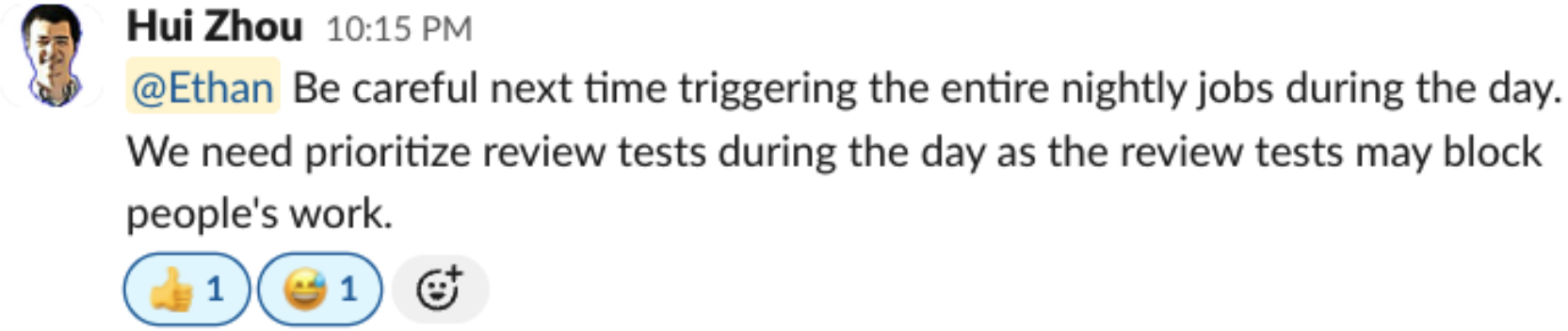
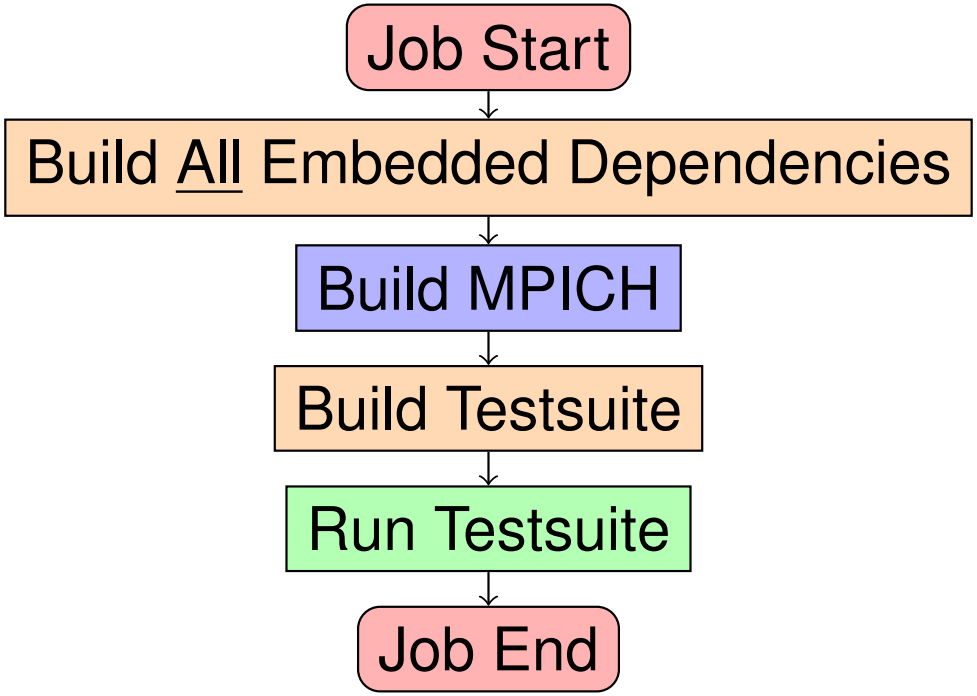


Figure: Re-running Nightly Testsuite at 2 PM

Qualms

Per-Run Rebuilds



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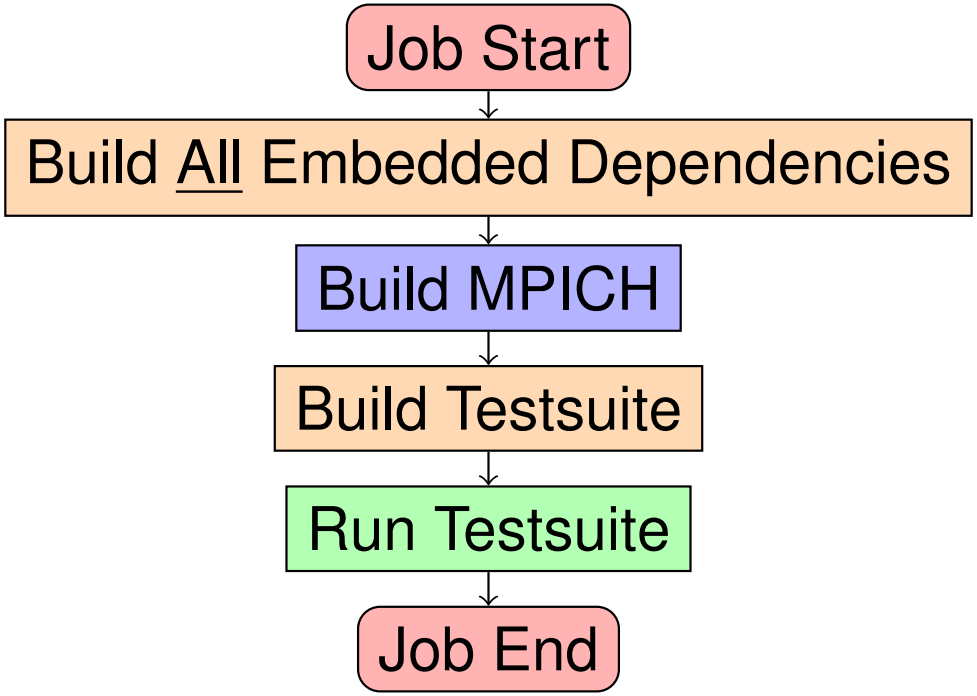

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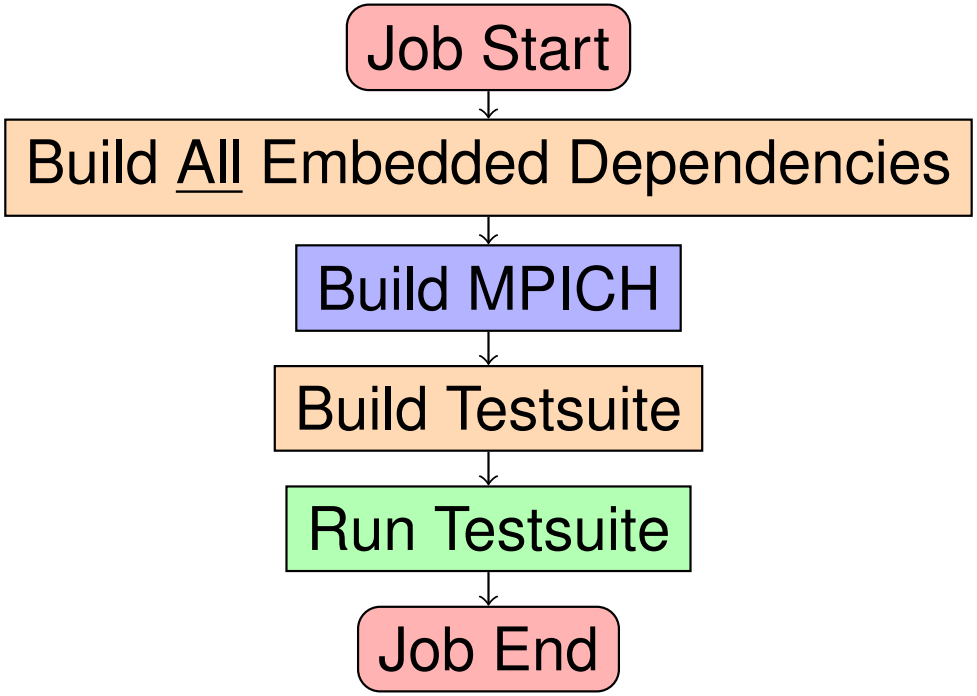

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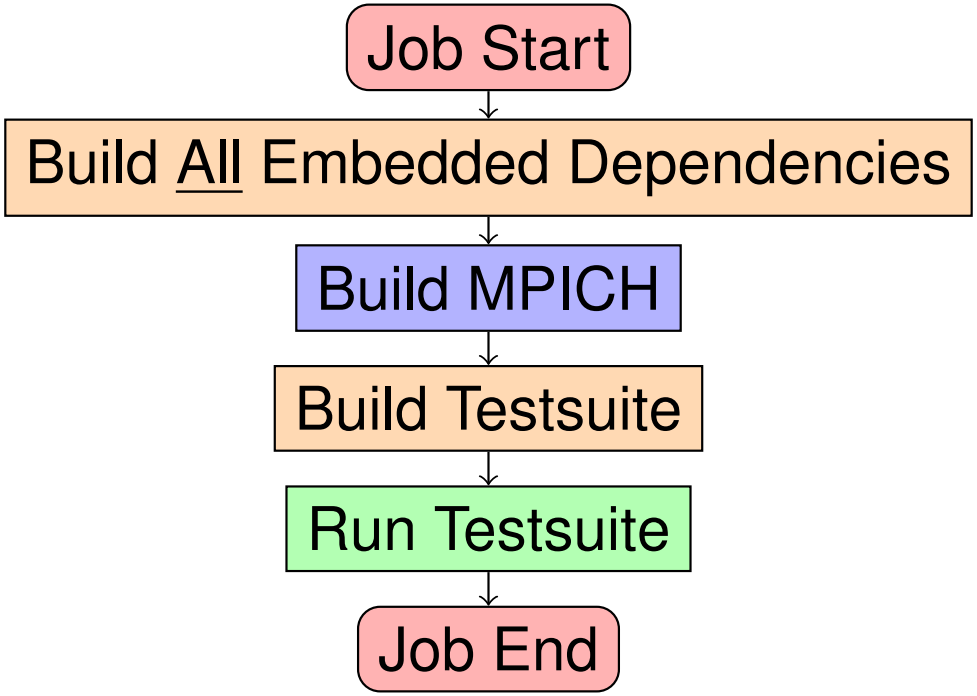

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3. HPC a first-class citizen

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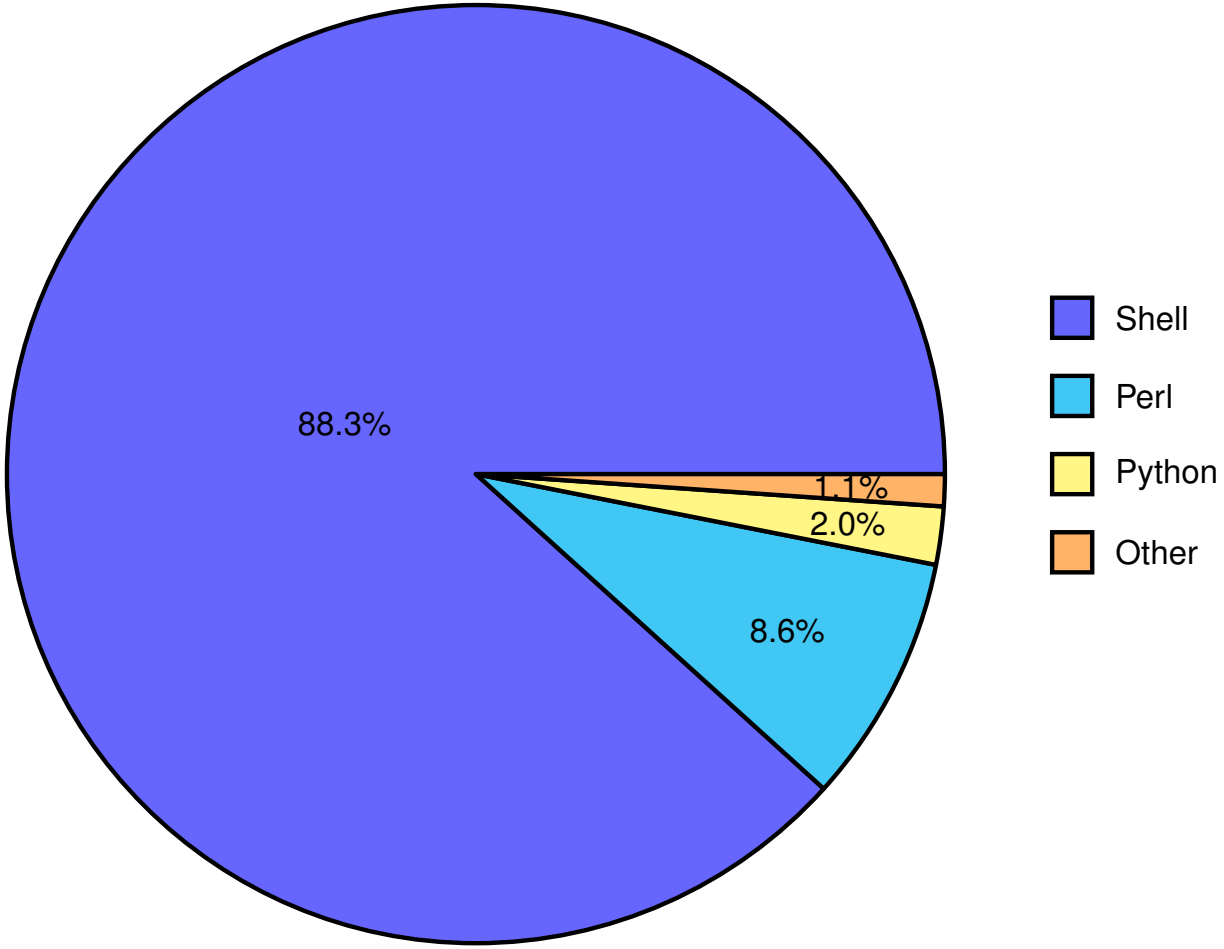

Figure: Jenkins Scripts Repository

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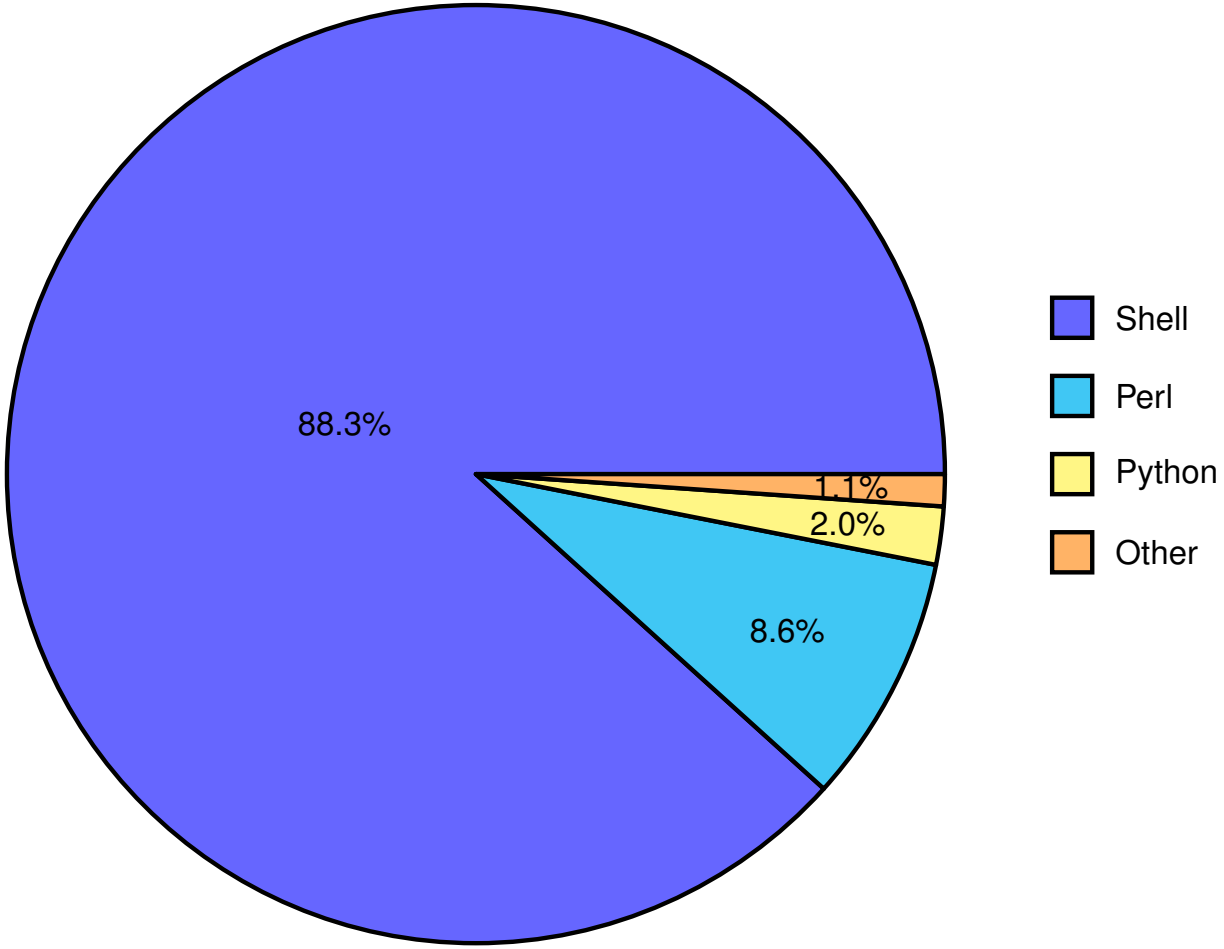

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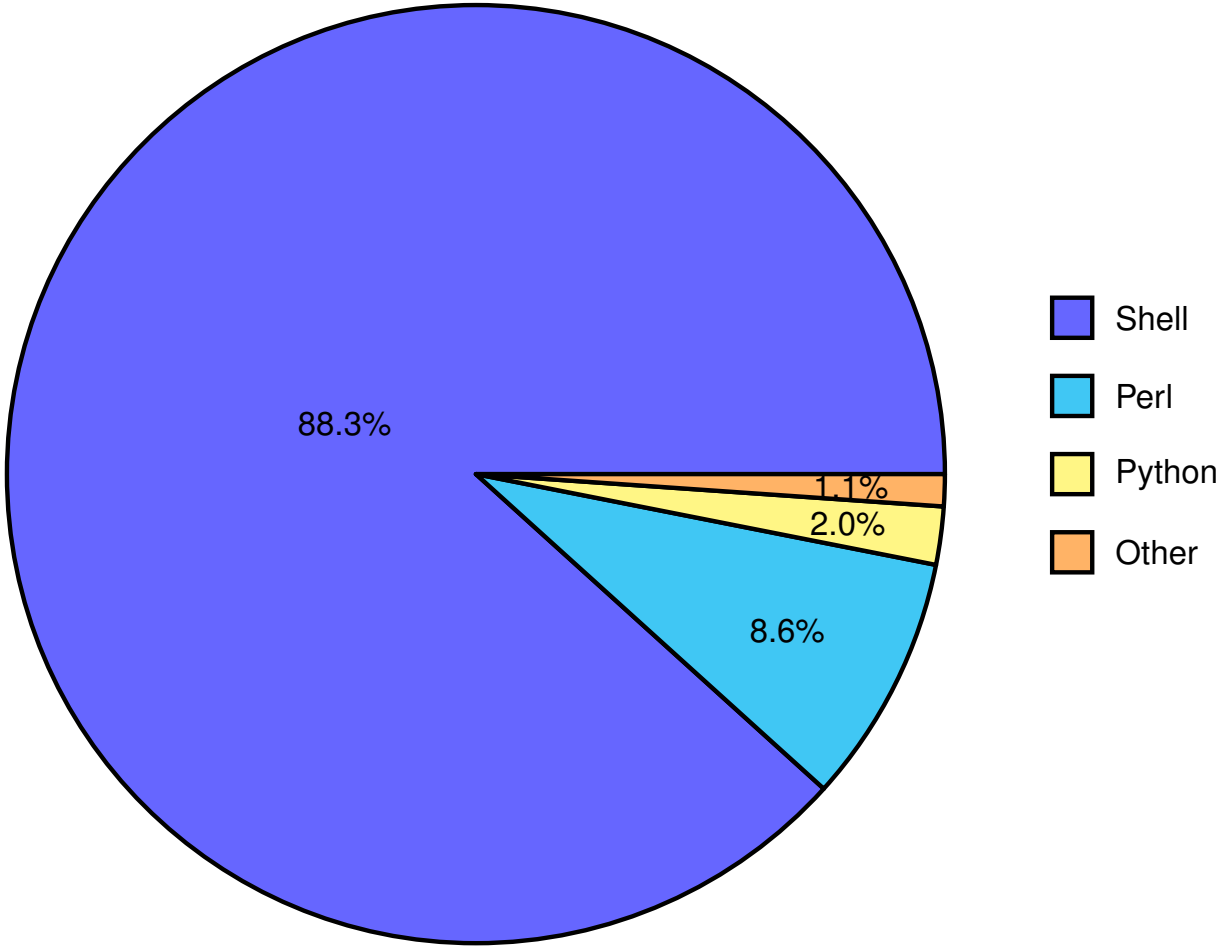

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Minimal changes $\Longrightarrow$ ***Substitute* compilation with Spack!**

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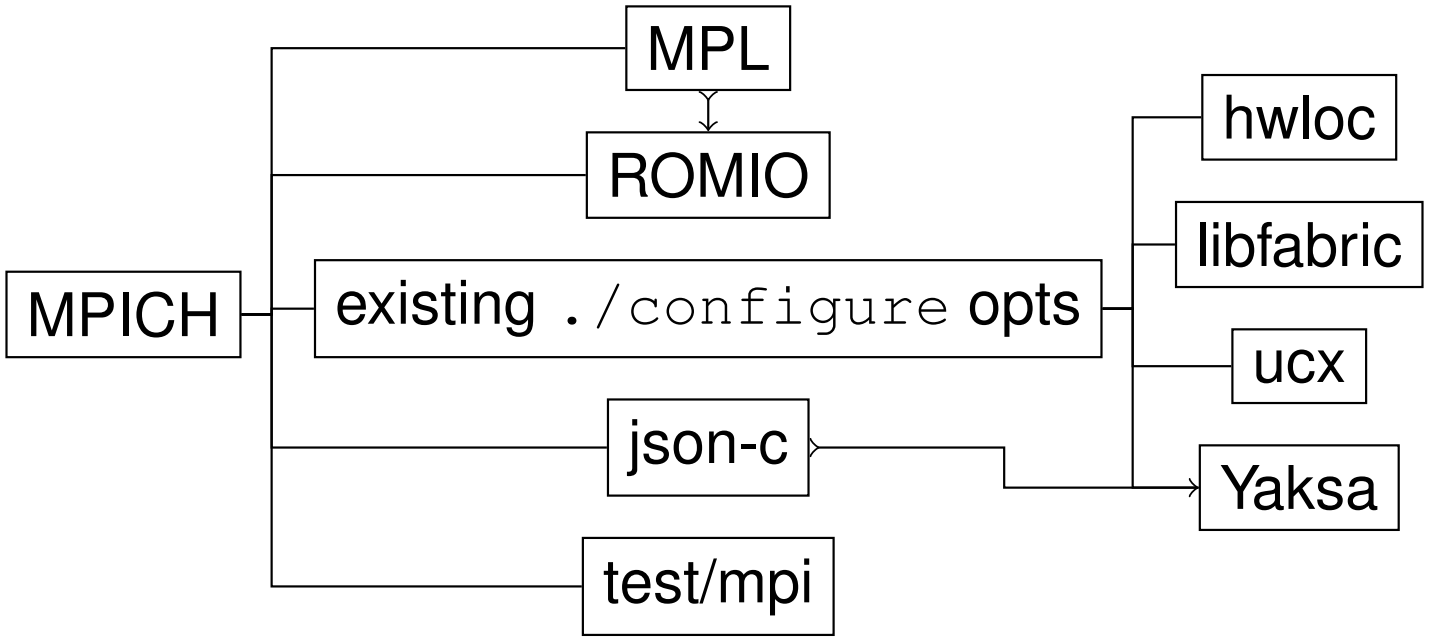

Figure: MPICH Dependency Graph[2]

[2] direct dependents

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Examples

```
from spack_repo.builtin.packages.ucx.package import Ucx as BuiltinUcx
from spack.package import *


class Ucx(BuiltinUcx):
        version(
                "1.19.0-embedded",
                git="https://github.com/pmodels/ucx.git",
                commit="4020d2be2271da0ac105d824f26ec76ad533577a",
        )

        @when("@1.19.0-embedded")
        def autoreconf(self, spec, prefix):
                Executable("./autogen.sh")()
```

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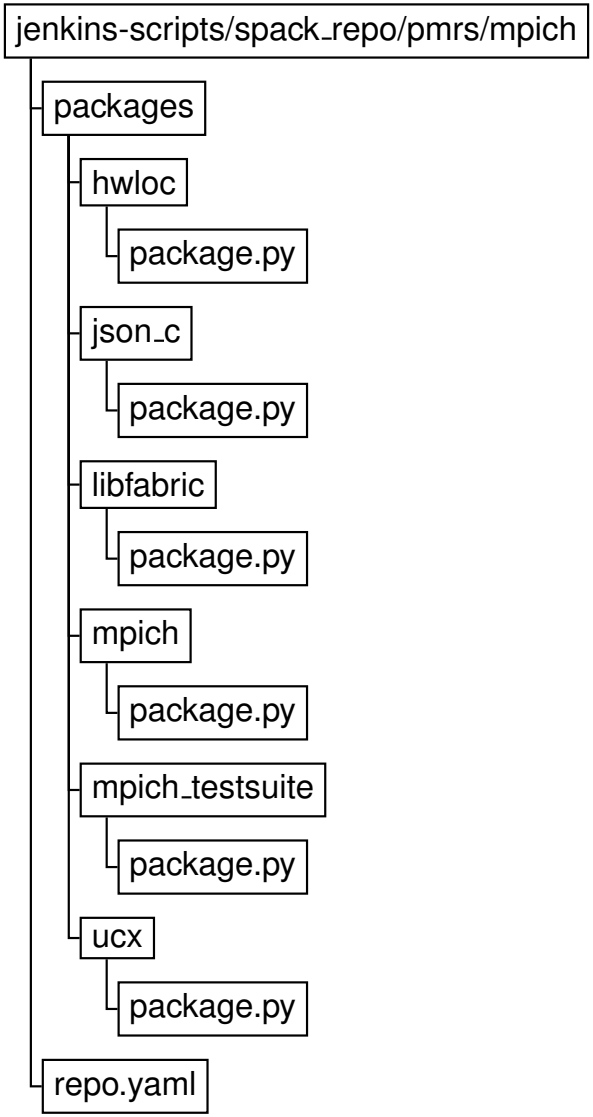

Figure: Internal PMRS Spack Repository

Integration with Jenkins

Our matrix jobs have custom *labels* which correspond to `./configure` options.

Examples

Configuration Matrix	gnu	clang
`ch3-tcp`	✓	✓
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Figure: `mpich-main-osx`[4] CI Job

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How do we translate these to `spack install`s?

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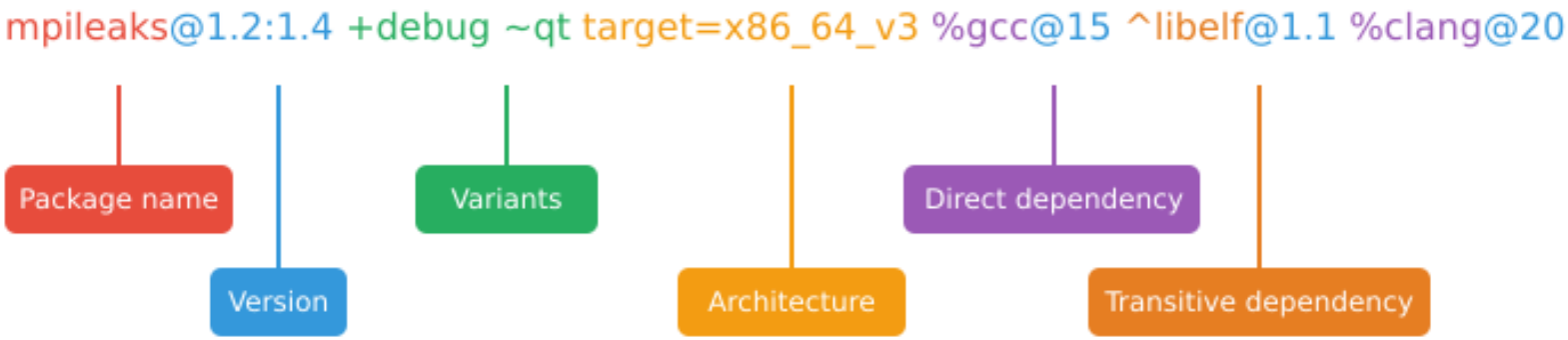

Figure: Spec Syntax[5]

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Let's generate equivalent Spack specs!

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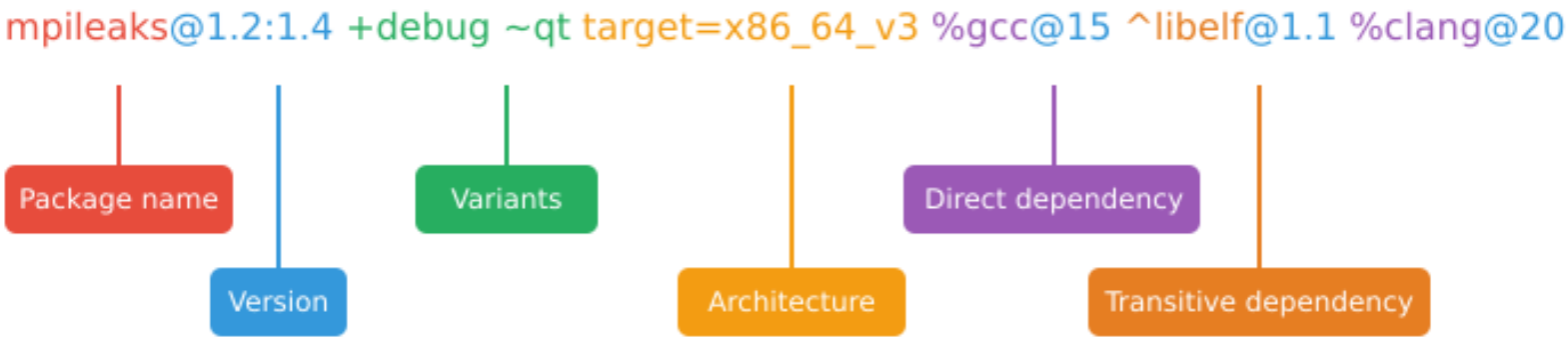

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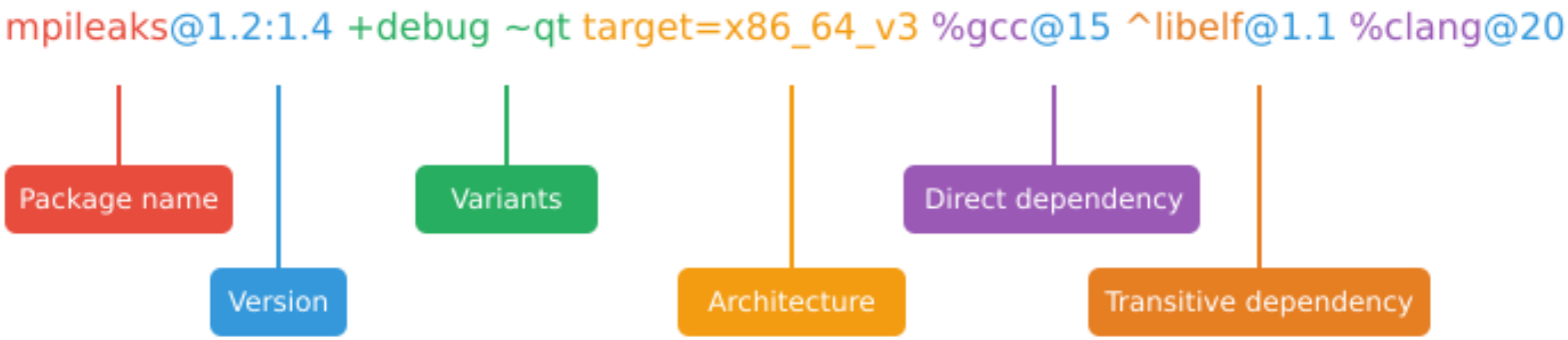

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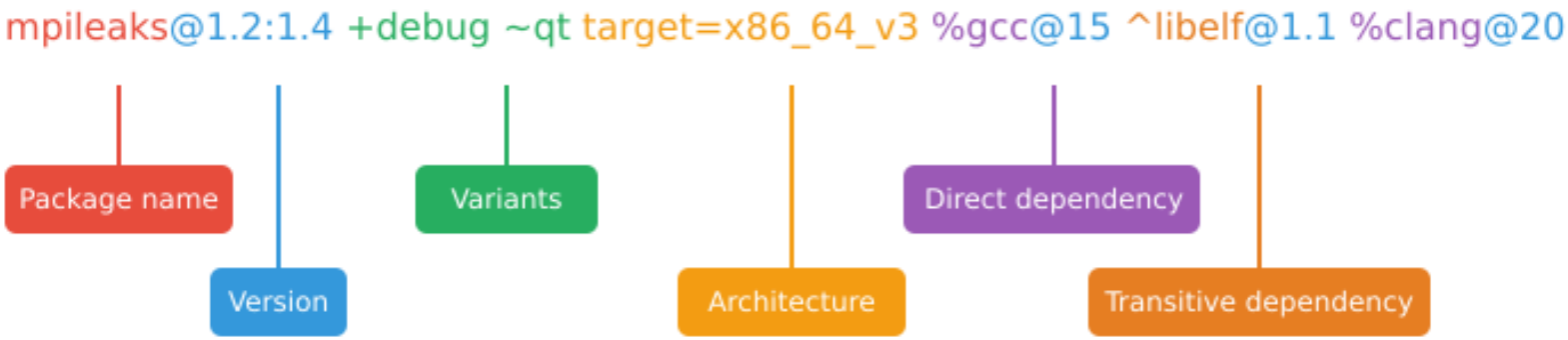

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Examples

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config_opt=' --with-device=ch4:ofi --without-xpmem
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$$\Downarrow$$

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Idea++

We can generalize this to a tool which templates Spack specs/environments for our CI[6] to concretize and install.

[6] and for us maintainers

PR Jobs

We can also utilize `spack develop` and `@<sha-1>` Git versions to validate contributed PRs *while utilizing Spack*!

[7] `jenkins-pmrs.cels.anl.gov/job/mpich-review-ch4-ofi-spack`

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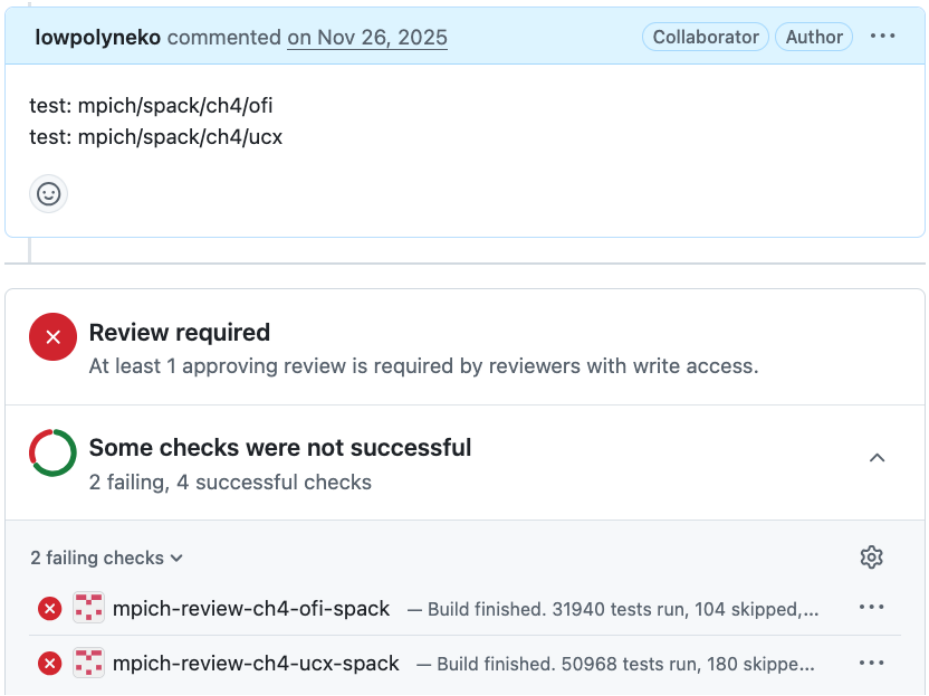

Figure: Example PR Utilizing Spackified CI Jobs[7,8]

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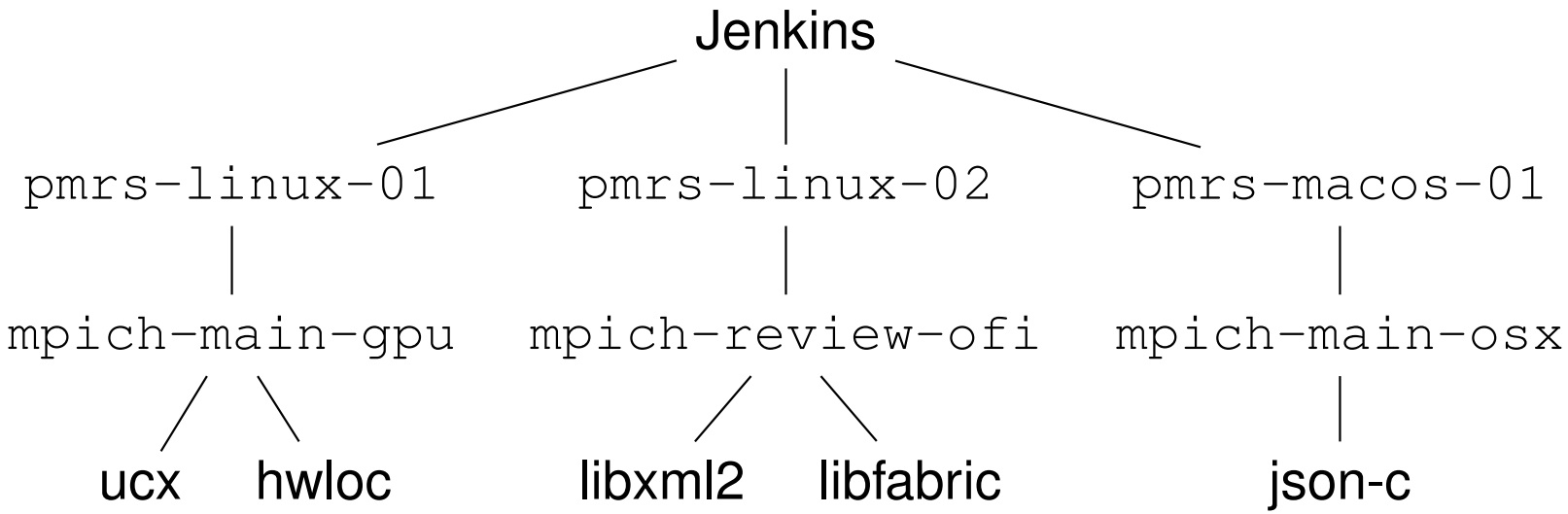

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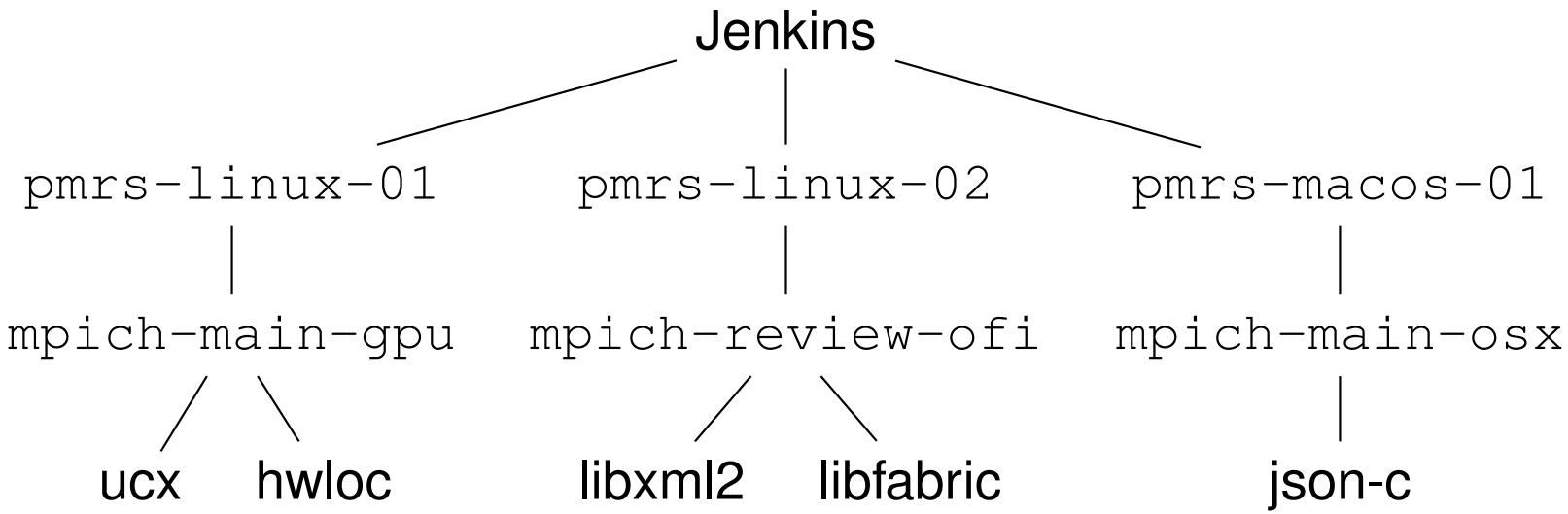

Bug!

We ran into a race condition with Spack's installer...

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bugfix: don't stop early when other spacks are installing #51539

Merged haampie merged 1 commit into `develop` from `bugfix-early-stop-on-multi-node-install` on Nov 10, 2025

Conversation 4 | Commits 1 | Checks 30 | Files changed 1

tgamblin commented on Nov 8, 2025 • edited — Member

Fixes #50933.

The installer in #47590 missed a loop condition and wasn't properly waiting on tasks requeued as `BuildStatus.INSTALLING`. This means another Spack is working on that task, and we should wait and not bail early.

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Thank you, Todd!

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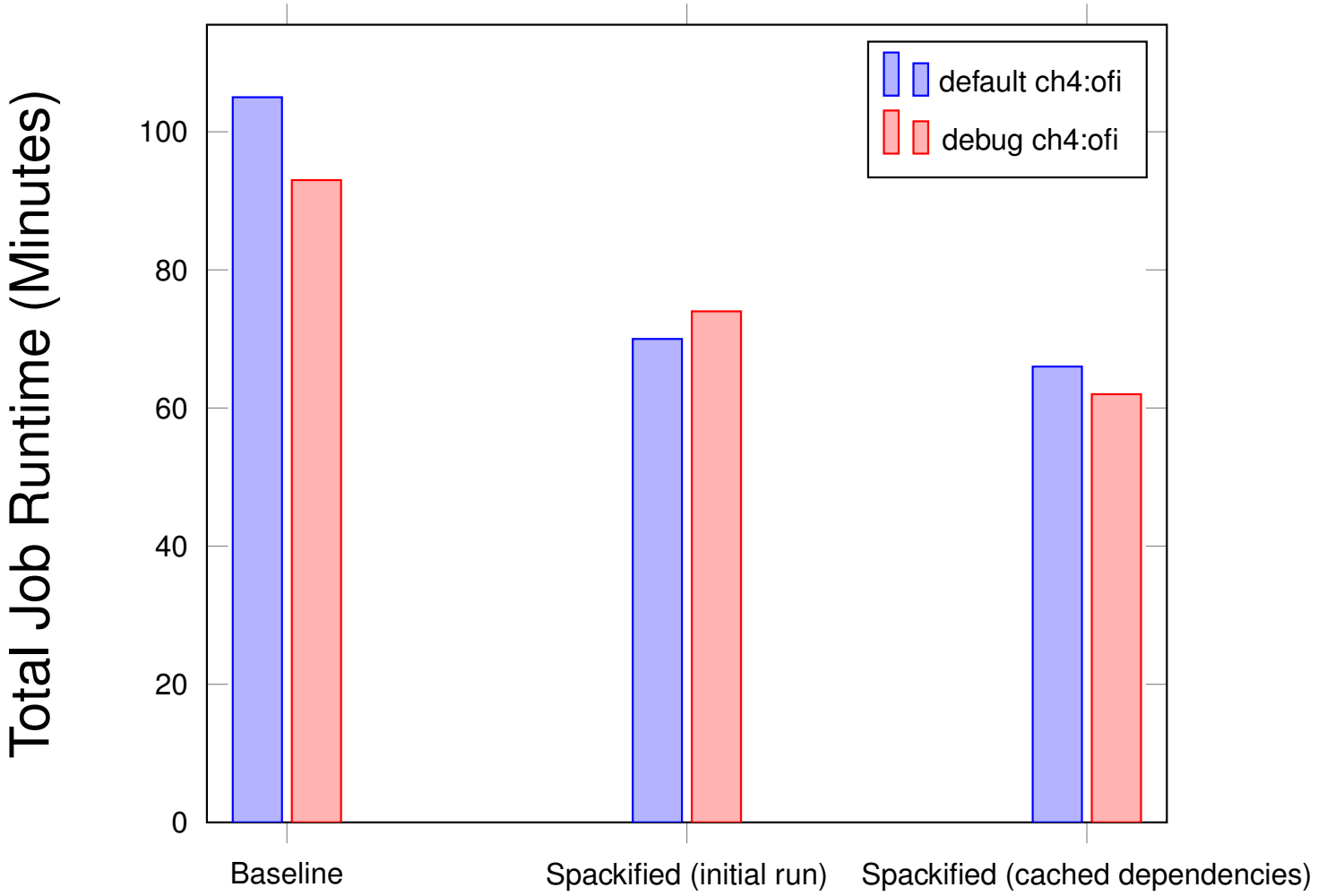

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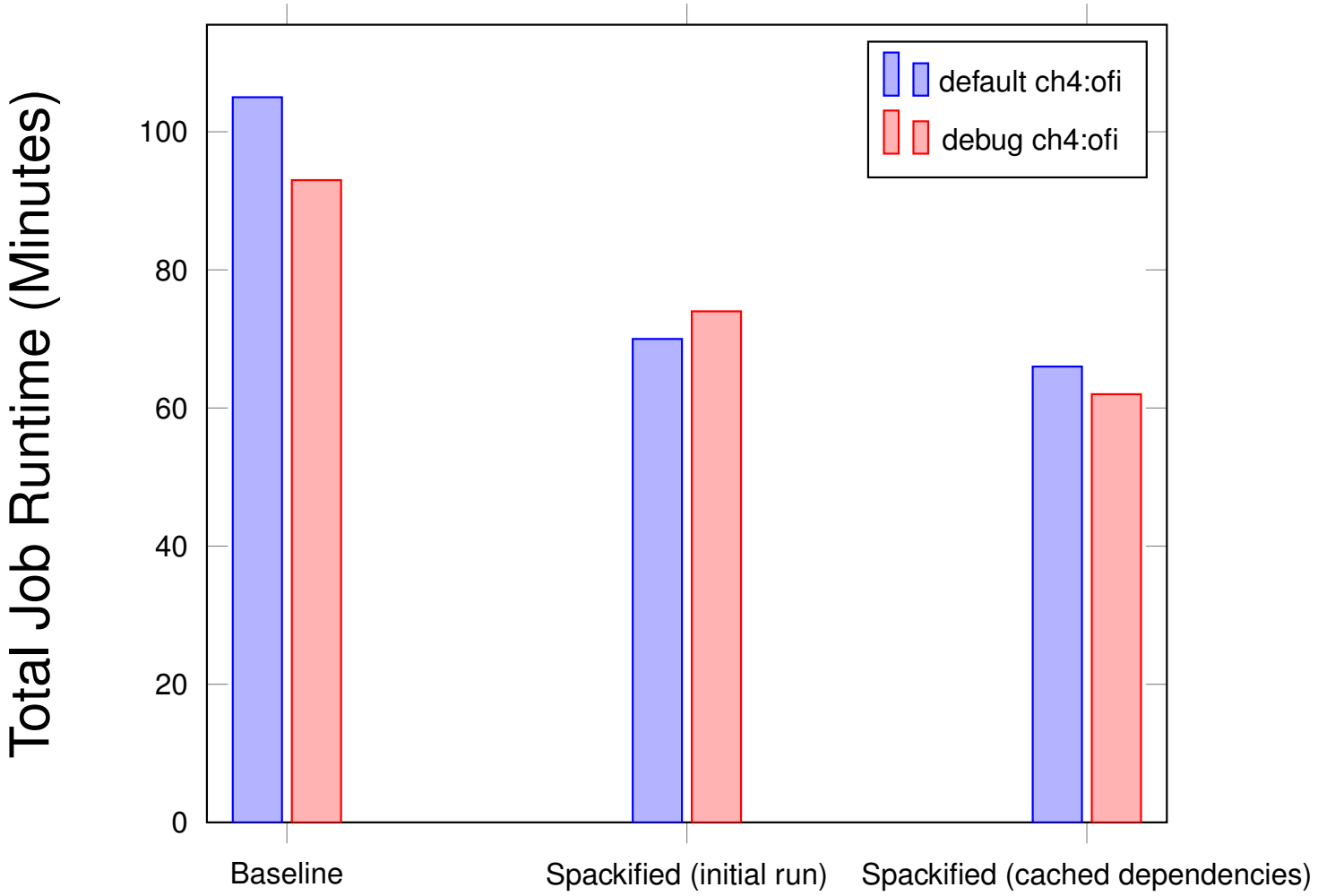

The primary bottleneck now is our single-threaded testsuite.

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- External dependencies now build independently of MPICH
- For *matrix configurations*, we now cache identical builds

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- Exporting test environment as OCI images?

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We see huge potential in further integration of Spack for our CI.

Our Ideas

- Refactoring internal, embedded modules to be cachable by Spack
  - MPL
  - ROMIO
  - etc...
- Pre-compilation of our entire testsuite
- Try integrating `ccache` for better compilation granularity
- Mirroring of CI builds for public usage
- Exporting test environment as OCI images?

The future is bright for our CI! ˆwˆ

Recap

Background
- MPICH
- Our Maintainer Workflow

Methodology
- Design/Integration
- Bug!

Results

Closing Remarks
- Future Work

Closing Remarks

Thank you!

- Hopefully this inspires other projects to attempt a similar endeavor.
- *We'll stick around to chat afterwards...*